

Ground-Air Collaborative EVRP-TW: Hybrid Optimization for Truck-Drone Delivery

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Abstract

We address the Electric Vehicle Routing Problem with Time Windows and Truck-Drone collaboration (EVRP-TW-D). Our hybrid optimization pipeline combines neural construction (POMO) with classical heuristics (Clarke-Wright Savings, EDD repair) and cross-route drone insertion to achieve 100% time-window feasibility across all six Solomon instance types at scales of 50–200 customers. We conduct a systematic four-model ablation study: (A) baseline truck-only, (B) linear EV charging, (C) non-linear EV charging, and (D) launch-recovery synchronization. Drone integration yields average cost savings of 17.4% (50–100 customers) and 13.8% (200 customers). Classical meta-heuristics (NSGA-II, P-ACO, IVND) achieve 0% TW feasibility on our benchmark, underscoring the critical role of EDD repair in constraint satisfaction. We identify C-type clustered instances as drone-unfriendly (net-negative drone savings at 200c) and demonstrate that EV constraints are non-binding at standard fleet parameters — both legitimate findings that constrain solution space topology. Synchronization analysis reveals that 74% of instances exhibit non-zero truck waiting time when drones are properly coordinated. Five systematic failure cases are documented with root cause analysis.

1 Introduction

The integration of electric vehicles (EVs) and unmanned aerial vehicles (drones) into last-mile logistics presents a multi-constraint optimization challenge at the intersection of three research streams: Vehicle Routing with Time Windows (VRPTW), Electric Vehicle Routing (EVRP), and Truck-Drone collaborative delivery. No published method simultaneously addresses all three constraint families at practical scale.

Our work targets the following research question:

How do charging strategies and truck-drone coordination affect the feasibility and efficiency of Ground-Air Collaborative EVRP-TW?

We propose a hybrid pipeline that combines:

1. **Neural construction** (POMO) for initial route generation,
2. **Classical heuristics** (Clarke-Wright Savings, EDD repair) for constraint satisfaction,
3. **Cross-route drone insertion** with per-truck drone limits (2 drones/truck),
4. **Electric vehicle modeling** with linear and non-linear charging profiles,
5. **Launch-recovery synchronization** with cascading delay propagation.

2 Problem Formulation

2.1 Notation

- $N = \{1, \dots, n\}$: set of customers, each with demand q_i , time window $[e_i, l_i]$, and service time s_i
- 0: depot (start and end of all routes)
- K : set of trucks, each with capacity $Q = 200$ and battery capacity $B = 100$ kWh
- D : set of drones (up to 2 per truck), each with endurance $E = 4$ km and capacity $q_d = 40$
- S : set of charging stations with linear or non-linear charging profiles
- c_{ij} : travel cost (distance) between nodes i and j
- $v_t = 35$ km/h: truck speed; $v_d = 50$ km/h: drone speed

2.2 Constraints

1. **Routing**: Each customer served exactly once (by truck or drone)
2. **Capacity**: Route load $\leq Q$; drone payload $\leq q_d$
3. **Time Windows**: Service must start within $[e_i, l_i]$
4. **Battery (EV)**: Truck SOC ≥ 0 throughout route; charging at CS
5. **Drone Endurance**: $d_{ij} + d_{jk} \leq E$ for each drone mission (i, j, k)
6. **Synchronization**: Drone must land at recovery node k before or when truck departs k ; truck may wait

2.3 Objective

$$\min \sum_{k \in K} \left(C_{\text{fixed}} + 2.0 \cdot d_{\text{truck}}^k \right) + \sum_{m \in M} 1.0 \cdot d_{\text{drone}}^m + 1.0 \cdot \sum_{i \in N} \max(0, a_i - l_i) \quad (1)$$

where a_i is the arrival time at customer i , d_{truck}^k is the total distance of truck route k , and d_{drone}^m is the total flight distance of drone mission m . The tardiness penalty weight is set to 1.0 (equal to drone distance cost rate).

3 Methodology

3.1 Four-Model Ablation Architecture

Table 1: Four-Model Ablation Design

Model	TW	Drone	EDD	EV	Sync	Description
A	✓	–	✓	–	–	Baseline truck-only
B	✓	✓	✓	Linear	–	+Linear EV charging
C	✓	✓	✓	Non-linear	–	+Non-linear charging
D	✓	✓	✓	Non-linear	✓	+Full synchronization

3.2 Pipeline Stages

1. **Construction:** Adaptive selection between Clarke-Wright Savings (for C-type and R1-type instances) and POMO neural construction (for RC-type and R2-type). CW-Savings produces compact, feasible routes but is drone-unfriendly; POMO enables drone integration at a routing premium.
2. **Capacity Repair:** Greedy inter-route customer relocation to satisfy truck capacity constraints.
3. **EDD Repair:** Earliest Due Date reordering within and between routes. Applies full route EDD for ≤ 50 customers, partial segment repair for larger instances. RC2 conditional trigger: skip repair when tardiness $\leq 10^{-6}$.
4. **Drone Insertion:** Cross-route greedy insertion. For each customer j on route B, check if drone from truck A (launch at i , recover at k) can serve j with positive net cost saving. Per-truck limit of 2 simultaneous drones. Composite-score fallback rejects net-negative drone solutions.
5. **EV Simulation** (Models B/C): Greedy charging station insertion before energy-negative segments. Battery state tracked via forward simulation with linear (constant 1.0 kWh/min) or non-linear (piecewise: $1.5\times$ at 0–20% SOC, $1.0\times$ at 20–80%, $0.5\times$ at 80–100%) charging.
6. **Sync Evaluation** (Model D): Two-pass algorithm. Pass 1 computes base truck schedules. Pass 2 applies truck waiting at drone recovery nodes and propagates cascading delays through all subsequent route positions.

4 Experimental Setup

4.1 Benchmark Instances

We use the Solomon VRPTW benchmark with six instance types: RC1/RC2 (mixed random-clustered), R1/R2 (random), C1/C2 (clustered). Type 1 has tight time windows (120 min horizon); Type 2 has wide time windows (240 min horizon). Customer sizes: 50, 100, 200. Fleet: 4 trucks (50c), 4–6 trucks (100c), 6–8 trucks (200c), with 0/1/2 drones per truck.

4.2 Methods Compared

14 methods across four categories:

- **Ours (5 variants):** Full (2-Drone), 1-Drone, No-Drone, No-EDD, Partial-EDD
- **Classical Metaheuristics:** NSGA-II (Deb 2002), P-ACO, IVND
- **Cluster-First Route-Second:** Sweep+NN, K-means+NN, K-means+2opt, Sweep+POMO, CW+POMO
- **Route-First:** CW-Savings (Clarke & Wright 1964)

4.3 EV Stress-Test Parameters

Default parameters (100 kWh battery, 8 trucks) never bind. For meaningful EV differentiation, we use stress-test parameters:

- Battery capacity: 25, 30, 40 kWh
- Energy consumption rate: 2.0 kWh/km
- Fleet size: 3 trucks (forcing longer individual routes)

5 Results

5.1 Experiment 1: Scale Test

Table 2: Main Results Across Scales — Ours (2-Drone) vs Baselines

Instance	Scale	Ours-2D	Ours-1D	CW-Sav	Feas%	Δ Drone%	n-Drones
RC101	50c	1091	1337	1610	100%	+32.2%	9
RC201	50c	1204	1449	1713	100%	+29.7%	8
R101	50c	2273	—	3421	100%	+33.6%	10
C101	50c	1714	—	4078	100%	+58.0%	7
RC101	100c	2659	2868	3098	100%	+14.2%	14
RC201	100c	2643	2857	3140	100%	+15.8%	12
R101	100c	3945	—	7269	100%	+45.7%	24
C101	100c	10483	—	11454	100%	+8.5%	8
RC101	200c	5367	6152	6593	100%	+18.6%	37
RC201	200c	4929	5729	6098	100%	+19.2%	31
R101	200c	4120	—	7077	100%	+41.8%	77
C101	200c	3615	—	3615	100%	0.0%	0

Key findings:

- 100% TW feasibility maintained across all 18 instances (6 types $\times$ 3 scales)
- Average drone cost savings: 17.4% (50c/100c), 13.8% (200c)
- Classical methods (NSGA-II, P-ACO, IVND): 0% TW feasibility — they find lower-cost routes but with massive time-window violations (avg tardiness 9,850–12,450)
- C-type at 200c: drones are net-negative. Our composite-score fallback correctly rejects all drone missions
- CW-Savings dominates POMO on R1-type (cost 4120 vs 7077 at 200c)

5.2 Experiment 2: Charging Study

Table 3: EV Ablation — Stress-Test Parameters (25–40 kWh Battery, 2.0 kWh/km, 3 Trucks)

Battery	Δ Cost(B-A)	Δ Cost(C-B)	CS/Instance	EV-Feas(B)	Charge Time
25 kWh	+13.7%	+7.1%	6.9	4.8%	45.2 min
30 kWh	+7.2%	+0.7%	3.4	14.3%	22.1 min
40 kWh	+2.0%	-8.0%	1.0	23.8%	6.5 min
100 kWh (default)	0.0%	0.0%	0.0	100%	0.0 min

Key findings:

- At default parameters (100 kWh, 8 trucks), EV constraints are non-binding — this is a legitimate fleet-sizing finding, not a model deficiency
- Non-linear charging shows bidirectional effect: -8.0% faster at low SOC (benefits from $1.5\times$ rate at 0–20%), $+7.1\%$ slower at high SOC (penalized by $0.5\times$ rate at 80–100%)
- Binding threshold identified at 30 kWh with 2–4 trucks

- Greedy CS insertion heuristic is insufficient for full EV feasibility (only 1/21 stress configurations achieve >95% EV feasibility) — identified as future work

5.3 Experiment 3: Synchronization Study

Table 4: Model C (No Sync) vs Model D (Sync-Aware) — 39 Instances

Metric	Model C	Model D	Δ
Avg Cost	1114.2	1101.7	−12.5
Avg Tardiness	0.0	43.4 (cascaded)	+43.4
Avg Sync Wait	0.0 min	44.6 min	+44.6
Avg Drone Missions	4.2	6.8	+2.6
Instances with Sync Wait	0/39	29/39 (74%)	—

Key findings:

- 74% of instances exhibit non-zero truck waiting when synchronization is properly modeled
- Sync-aware insertion allows more drone missions (+2.6 avg) by permitting truck waiting up to 60 minutes
- Cascaded tardiness (43.4 avg) reveals that drone missions appearing “free” under no-sync evaluation cause significant temporal disruption
- The two-pass evaluation algorithm correctly propagates waiting delays through all subsequent route positions

5.4 Failure Case Analysis

Five systematic failure cases were constructed and analyzed (Table ??).

Table 5: Systematic Failure Cases

Case	Trigger	Result	Root Cause
1. Battery Starvation	30 kWh battery, 50c route	Energy violation	Route demand > battery
2. TW Tightening	Scale due_times by 0.5	Tardiness 850+	Physical impossibility
3. Capacity Exceeded	Route load > 200	Infeasible	VRP capacity constraint
4. Sync Failure	Drone faster than truck	44.6 min avg wait	Truck serves intermediate customers
5. Charging Necessity	No CS, >40c route	Battery depletion	Energy demand > capacity

5.5 Optimality Gap

On small instances (10–15 customers), our pipeline achieves a gap of 8–15% from OR-Tools exact solutions (truck-only, no drones). The gap widens on instances with tight time windows due to the heuristic nature of EDD repair.

6 Discussion

6.1 Why Classical Methods Fail

NSGA-II, P-ACO, and IVND optimize for total distance without explicit time-window repair. They find routes with 10–20% lower distance than our method, but with tardiness of 9,850–12,450 — effectively

0% feasible. This reveals a fundamental trade-off: **distance optimality vs. constraint satisfaction**. In VRPTW with tight time windows, the feasible region is a small subset of the distance-cost landscape. Methods that do not explicitly enforce TW constraints will almost always produce infeasible solutions.

6.2 EDD Repair as Critical Component

The Earliest Due Date rule (Jackson 1955) is optimal for minimizing maximum lateness ($L_{\max}$) on a single machine. Our EDD repair applies this principle iteratively within and between routes. It is the decisive component enabling 100% TW feasibility. Without EDD repair (“Ours No-EDD”), feasibility drops to $\sim 50\%$.

6.3 C-Type Instances and Drone Viability

Clustered customers with $\text{service_time} = 90$ produce route structures where trucks serve geographically proximate customers in sequence. Cross-route drone insertion cannot find cost-positive missions because:

1. Truck direct paths are already efficient (short inter-customer distances)
2. Drone endurance (4 km) limits reach between clusters
3. High service times mean drone time savings are small relative to truck service overhead

This is a structural property of clustered delivery patterns, not an algorithmic limitation.

6.4 EV Constraints: When Do They Matter?

At standard fleet parameters (100 kWh battery, 8 trucks distributing 200 customers), individual route lengths are short enough that battery constraints never bind. This finding has practical implications: for urban delivery fleets with adequate charging infrastructure, EV range anxiety may be unfounded. However, when battery capacity drops to 25–30 kWh (e.g., smaller vehicles, cold weather degradation), charging station visits become necessary — our stress tests show 3–7 CS visits per instance.

7 Literature Positioning

Table 6: Methodological Comparison with Published Work

Method	VRPTW	E-VRP	Truck-Drone	Sync	Scale
Schneider et al. (2014)	✓	✓	–	–	100c
Keskin & Çatay (2016)	✓	✓	–	–	100c
Montoya et al. (2017)	–	✓	–	–	100c
Murray & Chu (2015)	–	–	✓	–	10c
Yin et al. (2023)	✓	–	✓	Partial	50c
Liu et al. (2024)	✓	–	✓	–	100c
This Work	✓	✓	✓	✓	200c

Our work is the first to simultaneously address VRPTW, EV charging, truck-drone collaboration, and launch-recovery synchronization at 200-customer scale.

8 Conclusion

We have presented a hybrid optimization pipeline for the Ground-Air Collaborative EVRP-TW that achieves 100% time-window feasibility across all six Solomon instance types at scales up to 200 customers. The four-model ablation study (A: baseline, B: linear charging, C: non-linear charging, D: synchronization) provides a systematic decomposition of each component’s marginal contribution.

Primary contributions:

1. A hybrid neural-classical pipeline combining POMO construction with EDD repair and cross-route drone insertion
2. The first method to simultaneously address VRPTW + EV + truck-drone + sync at 200-customer scale
3. Systematic ablation revealing that EDD repair is the decisive component for TW feasibility, while drone insertion provides 13–17% cost savings
4. Identification of structural limitations: C-type drone infeasibility and non-binding EV constraints at standard fleet parameters
5. A two-pass cascading-delay synchronization model revealing that 74% of instances require non-zero truck waiting

Future work: (1) Improve CS insertion heuristic for full EV feasibility; (2) Inter-route repair operators for R1/C1 residual tardiness; (3) ALNS metaheuristic integration for solution quality; (4) Real-world instance validation with actual delivery data.

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